

DATA SCIENCE PROGRAMING

LAB DA-4 CSI-3004

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1) QUESTION:

Assessment 4:

Tweet and Retweet Data Analysis Using R

Case Study: Identifying the Spread of Fake News on Social Media

A social-media analytics company wants to investigate how fake and genuine news propagate through social media. The company has collected tweet-level information containing the tweet ID, user, date, tweet text, number of retweets, likes, replies, and a `Fake/Real` label.

Using the R programming language, analyze the dataset and identify the characteristics and spread pattern of fake-labelled information.

Perform tweet and retweet data analysis using R to identify the spread pattern of fake news.

You are provided with a dataset containing the following attributes:

Attribute	Description
<code>Tweet_ID</code>	Unique identifier of the tweet
<code>User</code>	User who posted the tweet
<code>Date</code>	Date of posting
<code>Text</code>	Tweet content
<code>Retweets</code>	Number of retweets
<code>Likes</code>	Number of likes
<code>Replies</code>	Number of replies
<code>Label</code>	Fake or Real

TASKS:

1)

1. Data Creation and Exploration – 10 Marks

- Create/load the dataset in R.
- Display the first 10 records.
- Display the structure and summary statistics.
- Determine the number of Fake and Real tweets.

A) Create/load the dataset in R.:

CODE:

```
tweets <- read.csv(file.choose(), stringsAsFactors = FALSE)
|
tweets
```

OUTPUT:

	Retweets	Likes	Replies	Label
1	583	1053	117	Real
2	1425	3113	563	Fake
3	85	172	50	Real
4	1272	188	579	Fake
5	879	952	232	Real
6	1594	3380	168	Fake
7	589	686	113	Real
8	229	1636	104	Fake
9	561	227	376	Real
10	815	402	570	Fake
11	760	2414	101	Real
12	1394	1013	301	Fake
13	226	1606	145	Real
14	373	1596	368	Fake
15	166	750	276	Real
16	986	1634	281	Fake
17	469	1378	434	Real
18	509	3446	37	Fake
19	568	321	111	Real
20	684	950	516	Fake
21	959	635	138	Real
22	1189	2287	274	Fake

50	958	654	437	Fake
51	531	358	229	Real
52	240	287	558	Fake
53	504	731	211	Real
54	861	320	173	Fake
55	563	1913	149	Real
56	1644	2356	503	Fake
57	465	289	299	Real
58	1571	1364	63	Fake
59	1049	2225	83	Real
60	1783	841	75	Fake
61	501	1703	64	Real
62	1225	2515	45	Fake
63	1090	1345	136	Real
64	683	1057	276	Fake
65	634	1922	164	Real
66	59	1957	581	Fake
67	456	2122	138	Real
68	180	1080	383	Fake
69	639	2216	7	Real
70	653	2797	111	Fake
71	256	488	383	Real
72	617	2557	220	Fake
73	560	2120	253	Real
74	229	2678	438	Fake
75	27	1416	397	Real

B)VDisplay the first 10 records.

CODE:

```
head(tweets, 10)|
```

OUTPUT:

```
> head(tweets, 10)
```

	Tweet_ID	User	Date	Text	Retweets	Likes	Replies	Label
1	T0001	Socialwire	2026-01-08	official report confirms the latest development	583	1053	117	Real
2	T0002	Infoworld	2026-02-17	Breaking claim spreads rapidly across social media	1425	3113	563	Fake
3	T0003	Dailyupdate	2026-02-07	officials publish an update with verified information	85	172	50	Real
4	T0004	Trendwatch	2026-01-15	A misleading claim gains attention and rapid sharing	1272	188	579	Fake
5	T0005	Trendwatch	2026-02-15	Public records provide confirmed information about the development	879	952	232	Real
6	T0006	MediaPulse	2026-01-18	Breaking claim spreads rapidly across social media	1594	3380	168	Fake
7	T0007	BreakingNow	2026-01-28	New report provides confirmed details about the event	589	686	113	Real
8	T0008	NewsPoint	2026-01-22	Breaking claim spreads rapidly across social media	229	1636	104	Fake
9	T0009	TruthDesk	2026-01-24	New report provides confirmed details about the event	561	227	376	Real
10	T0010	GlobalReport	2026-02-04	Breaking claim spreads rapidly across social media	815	402	570	Fake

C) Display the structure and summary statistics.

CODE:

```
str(tweets)

# Summary statistics

summary(tweets)
```

OUTPUT:

```
> str(tweets)
'data.frame': 500 obs. of 8 variables:
 $ Tweet_ID: chr "T0001" "T0002" "T0003" "T0004" ...
 $ User : chr "SocialWire" "Infoworld" "DailyUpdate" "Trendwatch" ...
 $ Date : chr "2026-01-08" "2026-02-17" "2026-02-07" "2026-01-15" ...
 $ Text : chr "official report confirms the latest development" "Breaking claim spreads rapidly across social media" "officials publis
h an update with verified information" "A misleading claim gains attention and rapid sharing" ...
 $ Retweets: int 583 1425 85 1272 879 1594 589 229 561 815 ...
 $ Likes : int 1053 3113 172 188 952 3380 686 1636 227 402 ...
 $ Replies : int 117 563 50 579 232 168 113 104 376 570 ...
 $ Label : chr "Real" "Fake" "Real" "Fake" ...
>
> summary(tweets)
      Tweet_ID      User      Date      Text      Retweets      Likes      Replies
Length:500      Length:500      Length:500      Length:500      Min. : 25.0      Min. : 60      Min. : 5.0
Class :character Class :character Class :character Class :character 1st Qu.: 387.2      1st Qu.: 741      1st Qu.:138.0
Mode :character  Mode :character  Mode :character  Mode :character Median : 711.5      Median :1504      Median :259.0
                                          Mean : 745.2      Mean :1544      Mean :267.9
                                          3rd Qu.:1009.5      3rd Qu.:2231      3rd Qu.:384.2
                                          Max. :1791.0      Max. :3499      Max. :600.0

      Label
Length:500
Class :character
Mode :character
```

D) Determine the number of Fake and Real tweets.

CODE:

```
# d. Number of Fake and Real tweets

table(tweets$Label)|

prop.table(table(tweets$Label)) * 100
```

OUTPUT:

```
> table(tweets$Label)

Fake Real
 250   250
>
>
> # Percentage of Fake and Real tweets
>
> prop.table(table(tweets$Label)) * 100

Fake Real
  50    50
```

2) Retweet Analysis

A) Calculate the total number of retweets for Fake and Real tweets.

CODE:

```
total_retweets <- aggregate(  
  Retweets ~ Label,  
  data = tweets,  
  FUN = sum  
)  
  
total_retweets
```

OUTPUT:

```
> total_retweets  
  Label Retweets  
1 Fake    233066  
2 Real    139520
```

B) Calculate the average number of retweets for each category.

CODE:

```
average_retweets <- aggregate(  
  Retweets ~ Label,  
  data = tweets,  
  FUN = mean  
)  
  
average_retweets
```

OUTPUT:

```
> average_retweets  
  Label Retweets  
1 Fake    932.264  
2 Real    558.080
```

C) Identify the tweet with the maximum number of retweets.

CODE:

```
max_retweet <- tweets[which.max(tweets$Retweets), ]

max_retweet

# Display Tweet ID and retweet count

cat("Tweet with maximum retweets:\n")
cat("Tweet ID:", max_retweet$Tweet_ID, "\n")
cat("User:", max_retweet$User, "\n")
cat("Retweets:", max_retweet$Retweets, "\n")
cat("Label:", max_retweet$Label, "\n")
```

OUTPUT:

```
> max_retweet
  Tweet_ID      User      Date      Text Retweets Likes Replies Label
250    T0250 GlobalReport 2026-01-29 Viral post makes an unverified claim about a major event    1791  1565    106   Fake
>
> # Display Tweet ID and retweet count
>
> cat("Tweet with maximum retweets:\n")
Tweet with maximum retweets:
> cat("Tweet ID:", max_retweet$Tweet_ID, "\n")
Tweet ID: T0250
> cat("User:", max_retweet$User, "\n")
User: GlobalReport
> cat("Retweets:", max_retweet$Retweets, "\n")
Retweets: 1791
> cat("Label:", max_retweet$Label, "\n")
Label: Fake
```

D) Determine whether Fake or Real tweets have higher average retweet activity.

CODE:

```
fake_average <- mean(
  tweets$Retweets[tweets$Label == "Fake"]
)

real_average <- mean(
  tweets$Retweets[tweets$Label == "Real"]
)

cat("Average Fake Retweets:", fake_average, "\n")
cat("Average Real Retweets:", real_average, "\n")

if (fake_average > real_average) {
  cat("Fake tweets have higher average retweet activity.\n")
} else if (real_average > fake_average) {
  cat("Real tweets have higher average retweet activity.\n")
} else {
  cat("Fake and Real tweets have equal average retweet activity.\n")
}
```

OUTPUT:

```
>
> cat("Average Fake Retweets:", fake_average, "\n")
Average Fake Retweets: 932.264
> cat("Average Real Retweets:", real_average, "\n")
Average Real Retweets: 558.08
>
>
> if (fake_average > real_average) {
+   cat("Fake tweets have higher average retweet activity.\n")
+ } else if (real_average > fake_average) {
+   cat("Real tweets have higher average retweet activity.\n")
+ } else {
+   cat("Fake and Real tweets have equal average retweet activity.\n")
+ }
Fake tweets have higher average retweet activity.
```

3) Engagement Analysis:

A) Calculate the engagement score for every tweet.

CODE:

```
tweets$Engagement <- tweets$Retweets +
  tweets$Likes +
  tweets$Replies

# Display tweets with engagement score

tweets|
```

OUTPUT:

	Tweet_ID	User	Date	Text	Retweets	Likes	Replies	Label
1	T0001	SocialWire	2026-01-08	official report confirms the latest development	583	1053	117	Real
2	T0002	Infoworld	2026-02-17	Breaking claim spreads rapidly across social media	1425	3113	563	Fake
3	T0003	DailyUpdate	2026-02-07	officials publish an update with verified information	85	172	50	Real
4	T0004	Trendwatch	2026-01-15	A misleading claim gains attention and rapid sharing	1272	188	579	Fake
5	T0005	Trendwatch	2026-02-15	Public records provide confirmed information about the development	879	952	232	Real
6	T0006	MediaPulse	2026-01-18	Breaking claim spreads rapidly across social media	1594	3380	168	Fake
7	T0007	BreakingNow	2026-01-28	New report provides confirmed details about the event	589	686	113	Real
8	T0008	NewsPoint	2026-01-22	Breaking claim spreads rapidly across social media	229	1636	104	Fake
9	T0009	TruthDesk	2026-02-24	New report provides confirmed details about the event	561	227	376	Real
10	T0010	GlobalReport	2026-02-04	Breaking claim spreads rapidly across social media	815	402	570	Fake
11	T0011	NewsToday	2026-02-23	Public records provide confirmed information about the development	760	2414	101	Real
12	T0012	BreakingNow	2026-01-05	Breaking claim spreads rapidly across social media	1394	1013	301	Fake
13	T0013	DailyUpdate	2026-02-24	verified information released by the authorities	226	1606	145	Real
14	T0014	GlobalReport	2026-02-10	Social media users share a shocking unconfirmed report	373	1596	368	Fake
15	T0015	Trendwatch	2026-02-12	New report provides confirmed details about the event	166	750	276	Real
16	T0016	BreakingNow	2026-01-16	Viral post makes an unverified claim about a major event	986	1634	281	Fake
17	T0017	WorldNews	2026-02-10	Public records provide confirmed information about the development	469	1378	434	Real
18	T0018	NewsPoint	2026-02-19	Breaking claim spreads rapidly across social media	509	3446	37	Fake
19	T0019	NewsPoint	2026-01-21	officials publish an update with verified information	568	321	111	Real
20	T0020	WorldNews	2026-02-06	viral message contains a false statement about current events	684	950	516	Fake
21	T0021	ViralNews	2026-02-26	Public records provide confirmed information about the development	959	635	138	Real
22	T0022	Infoworld	2026-01-16	Viral message contains a false statement about current events	1189	2287	274	Fake
23	T0023	BreakingNow	2026-02-07	officials publish an update with verified information	837	1532	115	Real
24	T0024	Infoworld	2026-02-02	unverified information becomes widely circulated online	226	3175	53	Fake
25	T0025	PublicWire	2026-01-08	verified information released by the authorities	313	1330	308	Real

59	T0059	NewsHub	2026-02-07	Officials publish an update with verified information	1049	2225	83	Real
60	T0060	NewsHub	2026-02-02	Breaking claim spreads rapidly across social media	1783	841	75	Fake
61	T0061	MediaPulse	2026-01-05	Public records provide confirmed information about the development	501	1703	64	Real
62	T0062	WorldNews	2026-02-06	Viral post makes an unverified claim about a major event	1225	2515	45	Fake
63	T0063	MediaPulse	2026-01-06	Officials publish an update with verified information	1090	1345	136	Real
64	T0064	Trendwatch	2026-02-12	Viral message contains a false statement about current events	683	1057	276	Fake
65	T0065	ViralNews	2026-01-09	Public records provide confirmed information about the development	634	1922	164	Real
66	T0066	WorldNews	2026-02-18	Breaking claim spreads rapidly across social media	59	1957	581	Fake
67	T0067	DailyUpdate	2026-01-05	Reliable sources confirm the latest news	456	2122	138	Real
68	T0068	Infoworld	2026-03-01	Social media users share a shocking unconfirmed report	180	1080	383	Fake
69	T0069	NewsToday	2026-01-11	Officials publish an update with verified information	639	2216	7	Real
70	T0070	SocialWire	2026-02-22	A misleading claim gains attention and rapid sharing	653	2797	111	Fake
71	T0071	WorldNews	2026-01-09	New report provides confirmed details about the event	256	488	383	Real
72	T0072	FactCheck	2026-01-10	Social media users share a shocking unconfirmed report	617	2557	220	Fake
73	T0073	BreakingNow	2026-01-22	Verified information released by the authorities	560	2120	253	Real
74	T0074	NewsToday	2026-02-27	Breaking claim spreads rapidly across social media	229	2678	438	Fake
75	T0075	PublicVoice	2026-01-18	Official report confirms the latest development	27	1416	397	Real
76	T0076	Infoworld	2026-02-10	Social media users share a shocking unconfirmed report	370	3116	457	Fake
77	T0077	Factcheck	2026-02-15	Officials publish an update with verified information	39	508	41	Real
78	T0078	WorldNews	2026-02-14	Viral post makes an unverified claim about a major event	1157	227	383	Fake
79	T0079	MediaPulse	2026-02-05	Verified information released by the authorities	900	572	24	Real
80	T0080	NewsToday	2026-01-24	Breaking claim spreads rapidly across social media	772	940	260	Fake
81	T0081	SocialWire	2026-01-07	New report provides confirmed details about the event	852	683	124	Real
82	T0082	PublicVoice	2026-01-11	Viral post makes an unverified claim about a major event	884	181	188	Fake
83	T0083	BreakingNow	2026-03-01	New report provides confirmed details about the event	863	1066	139	Real

Engagement

1	1753
2	5101
3	307
4	2039
5	2063
6	5142
7	1388
8	1969
9	1164
10	1787
11	3275
12	2708
13	1977
14	2337
15	1192
16	2901
17	2281
18	3992
19	1000
20	2150
21	1732
22	3750
23	2484

B) Calculate the average engagement for Fake and Real tweets.:

CODE:

```
average_engagement <- aggregate(  
  Engagement ~ Label,  
  data = tweets,  
  FUN = mean  
)  
  
average_engagement  
  
# Separate averages  
  
fake_engagement <- mean(  
  tweets$Engagement[tweets$Label == "Fake"]  
)  
  
real_engagement <- mean(  
  tweets$Engagement[tweets$Label == "Real"]  
)  
  
cat("Average Fake Engagement:", fake_engagement, "\n")  
cat("Average Real Engagement:", real_engagement, "\n")
```

OUTPUT:

```
> cat("Average Fake Engagement:", fake_engagement, "\n")  
Average Fake Engagement: 3019.96  
> cat("Average Real Engagement:", real_engagement, "\n")  
Average Real Engagement: 2093.772
```

C) Identify the top five tweets based on engagement.

CODE:

```
top_five <- tweets[  
  order(tweets$Engagement, decreasing = TRUE),  
]  
  
top_five <- head(top_five, 5)  
  
top_five  
  
# Display important columns  
  
top_five[, c(  
  "Tweet_ID",  
  "User",  
  "Label",  
  "Retweets",  
  "Likes",  
  "Replies",  
  "Engagement"  
)]
```

OUTPUT:

```
+ / J  
  Tweet_ID      User Label Retweets Likes Replies Engagement  
252    T0252 NewsToday Fake    1776  3455    470      5701  
136    T0136 Trendwatch Fake    1454  3482    253      5189  
176    T0176 Trendwatch Fake    1653  3320    207      5180  
212    T0212 DailyUpdate Fake    1785  2842    551      5178  
108    T0108 ViralNews Fake    1686  2930    535      5151  
> |
```

4) Temporal Spread Analysis;

A). Calculate the total number of retweets for Fake and Real tweets for each date.

CODE:

```
tweets$Date <- as.Date(tweets$Date)

# a. Total number of retweets for Fake and Real
# tweets for each date

daily_retweets <- aggregate(
  Retweets ~ Date + Label,
  data = tweets,
  FUN = sum
)

daily_retweets

# Create a complete date-label table if needed

daily_retweets
```

OUTPUT:

	Date	Label	Retweets
1	2026-01-01	Fake	2275
2	2026-01-03	Fake	2476
3	2026-01-04	Fake	1787
4	2026-01-05	Fake	8136
5	2026-01-06	Fake	3992
6	2026-01-07	Fake	3440
7	2026-01-08	Fake	578
8	2026-01-09	Fake	6672
9	2026-01-10	Fake	4841
10	2026-01-11	Fake	5409
11	2026-01-12	Fake	2901
12	2026-01-13	Fake	4200
13	2026-01-14	Fake	7344
14	2026-01-15	Fake	9434
15	2026-01-16	Fake	7429
16	2026-01-17	Fake	2053
17	2026-01-18	Fake	7475
18	2026-01-19	Fake	5032
19	2026-01-20	Fake	5384
20	2026-01-21	Fake	3425
21	2026-01-22	Fake	2672
22	2026-01-23	Fake	1639
23	2026-01-24	Fake	3032

B) Plot a line graph showing the spread of Fake and Real tweets over time.

CODE:

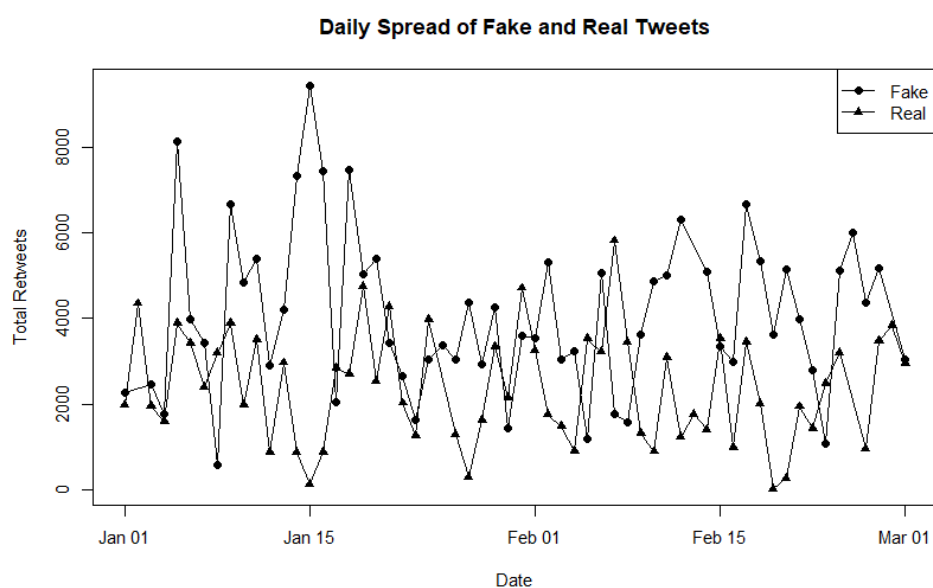
```
fake_daily <- daily_retweets[
  daily_retweets$Label == "Fake",
]

real_daily <- daily_retweets[
  daily_retweets$Label == "Real",
]

plot(
  fake_daily$Date,
  fake_daily$Retweets,|
  type = "o",
  pch = 16,
  xlab = "Date",
  ylab = "Total Retweets",
  main = "Daily Spread of Fake and Real Tweets",
  ylim = range(daily_retweets$Retweets)
)

lines(
  real_daily$Date,
  real_daily$Retweets,
  type = "o",
  pch = 17
)
```

OUTPUT:



C) Identify the date on which Fake-labelled tweets achieved their highest propagation.

CODE:

```
fake_max <- fake_daily[
  which.max(fake_daily$Retweets),
]

cat(
  "Date of highest Fake tweet propagation:",
  fake_max$Date, "\n"
)

cat(
  "Maximum Fake retweets:",
  fake_max$Retweets, "\n"
)
```

OUTPUT:

```
. /
Date of highest Fake tweet propagation: 20468
>
> cat(
+   "Maximum Fake retweets:",
+   fake_max$Retweets, "\n"
+ )
Maximum Fake retweets: 9434
```

D) . Explain the observed spread pattern.

CODE:

```
cat("\nSpread Pattern Analysis:\n")

if (fake_average > real_average) {
  cat(
    "Fake-labelled tweets have higher average retweet activity ",
    "than Real-labelled tweets, indicating faster propagation ",
    "of fake information.\n"
  )
} else {
  cat(
    "Real-labelled tweets have higher average retweet activity ",
    "than Fake-labelled tweets, indicating greater propagation ",
    "of genuine information.\n"
  )
}
```

OUTPUT:

```
Fake-labelled tweets have higher average retweet activity than Real-labelled tweets, indicating faster propagation of fake information.
```

5) Influential User Analysis:

Calculate the following for each user:

A) Number of tweets;

CODE:

```
user_retweets <- aggregate(  
  Retweets ~ User,  
  data = tweets,  
  FUN = sum  
)  
  
names(user_retweets)[2] <- "Total_Retweets"
```

B) Total retweets:

```
user_avg_retweets <- aggregate(  
  Retweets ~ User,  
  data = tweets,  
  FUN = mean  
)  
  
names(user_avg_retweets)[2] <- "Average_Retweets"
```

C) Average retweets:

```
user_avg_retweets <- aggregate(  
  Retweets ~ User,  
  data = tweets,  
  FUN = mean  
)  
  
names(user_avg_retweets)[2] <- "Average_Retweets"
```

D) Total engagement:

```
user_engagement <- aggregate(
  Engagement ~ User,
  data = tweets,
  FUN = sum
)

names(user_engagement)[2] <- "Total_Engagement"

# Combine all user statistics

user_analysis <- merge(
  user_tweets,
  user_retweets,
  by = "User"
)

user_analysis <- merge(
  user_analysis,
  user_avg_retweets,
  by = "User"
)

user_analysis <- merge(
  user_analysis,
  user_avg_retweets,
  by = "User"
)

user_analysis <- merge(
  user_analysis,
  user_engagement,
  by = "User"
)

# Display user analysis

user_analysis

# Sort users according to total retweets

user_analysis <- user_analysis[
  order(
    user_analysis$Total_Retweets,
    decreasing = TRUE
  ),
]
```

the top three users based on total retweets.

CODE:

```
top_three_users <- head(user_analysis, 3)
top_three_users

cat("\nTop Three Users Based on Total Retweets:\n")
print(
  top_three_users[
    ,
    c(
      "User",
      "Number_of_Tweets",
      "Total_Retweets",
      "Average_Retweets",
      "Total_Engagement"
    )
  ]
)
```

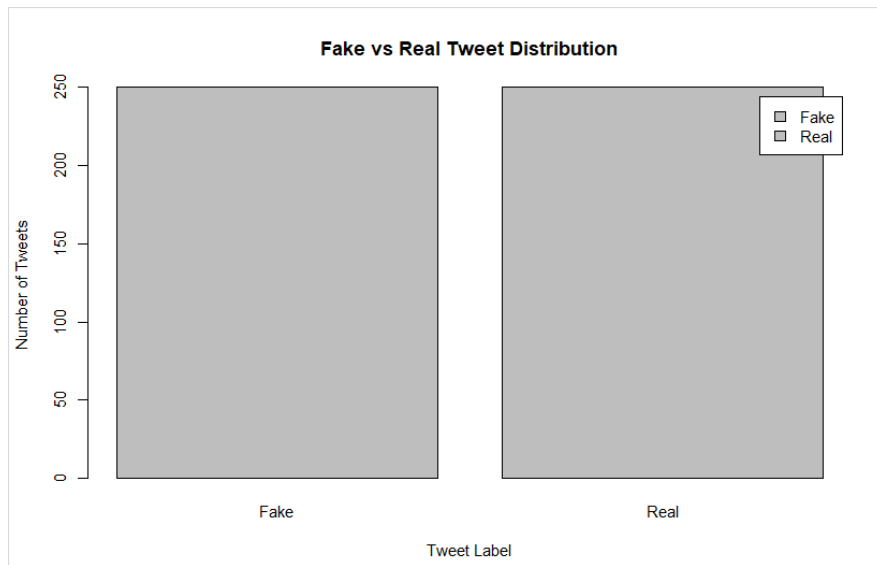
6) . Visualization:

A) Fake vs. Real tweet distribution;

```
tweet_distribution <- table(tweets$Label)

barplot(
  tweet_distribution,
  main = "Fake vs Real Tweet Distribution",
  xlab = "Tweet Label",
  ylab = "Number of Tweets",
  legend.text = TRUE
)
```

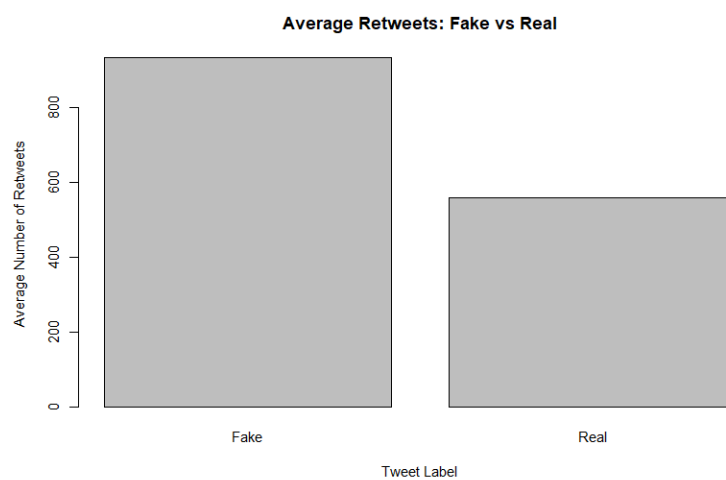

OUTPUT:



B) Retweet comparison:

```
retweet_values <- tapply(  
  tweets$Retweets,  
  tweets$Label,  
  mean  
)  
  
barplot(  
  retweet_values,  
  main = "Average Retweets: Fake vs Real",  
  xlab = "Tweet Label",  
  ylab = "Average Number of Retweets"  
)
```

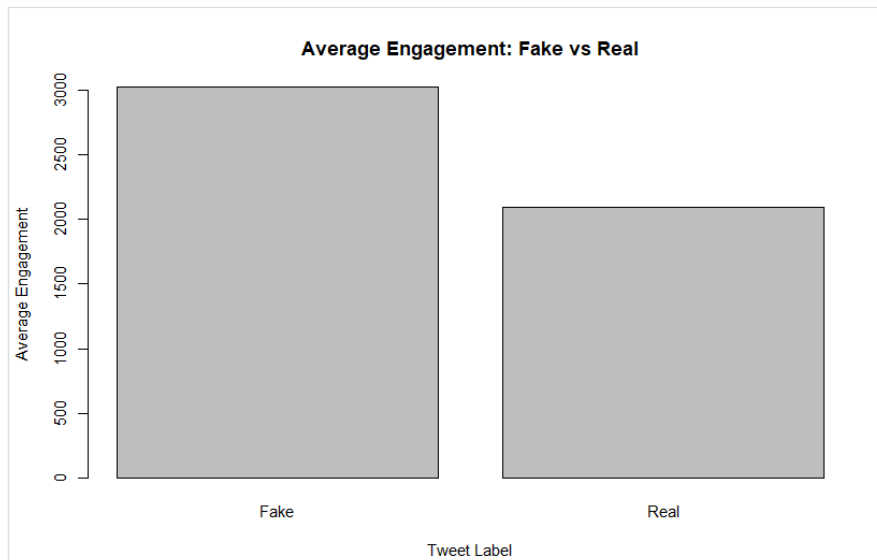
OUTPUT:



C) Engagement comparison;

```
engagement_values <- tapply(  
  tweets$Engagement,  
  tweets$Label,  
  mean  
)  
  
barplot(  
  engagement_values,  
  main = "Average Engagement: Fake vs Real",  
  xlab = "Tweet Label",  
  ylab = "Average Engagement"  
)
```

OUTPUT:



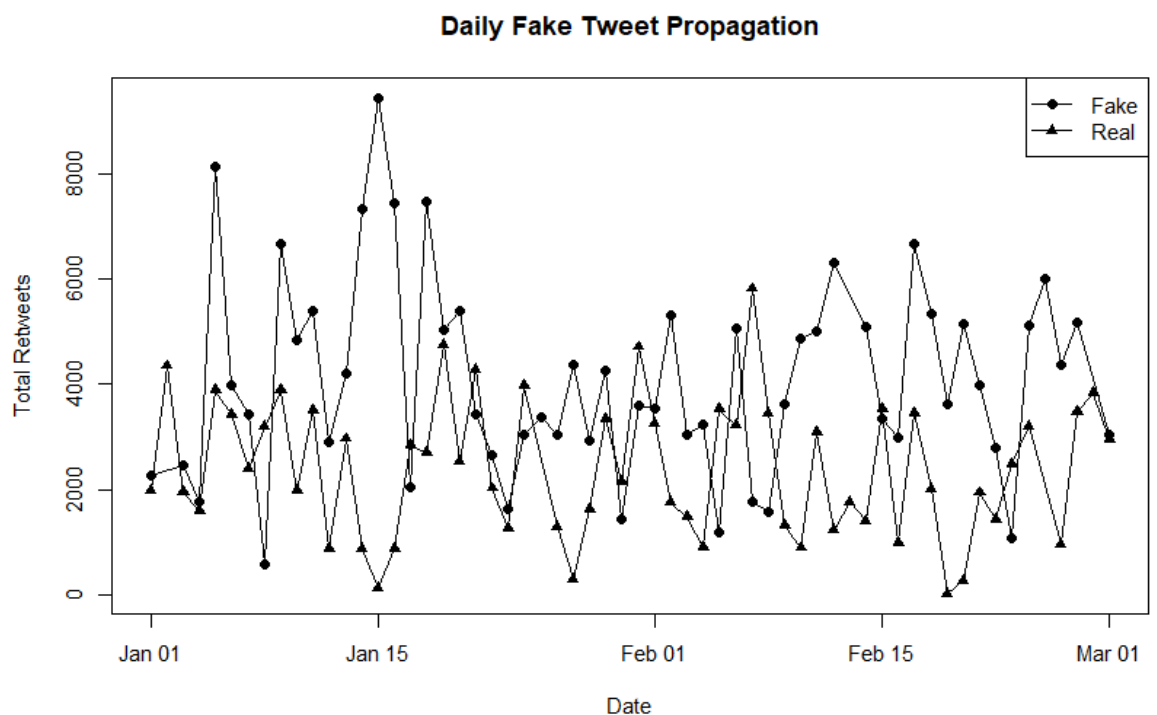
D) Daily spread pattern:

```
plot(
  fake_daily$Date,
  fake_daily$Retweets,
  type = "o",
  pch = 16,
  xlab = "Date",
  ylab = "Total Retweets",
  main = "Daily Fake Tweet Propagation",
  ylim = range(daily_retweets$Retweets)
)

lines(
  real_daily$Date,
  real_daily$Retweets,
  type = "o",
  pch = 17
)

legend(
  "topright",
  legend = c("Fake", "Real"),
  pch = c(16, 17),
  lty = 1
)
```

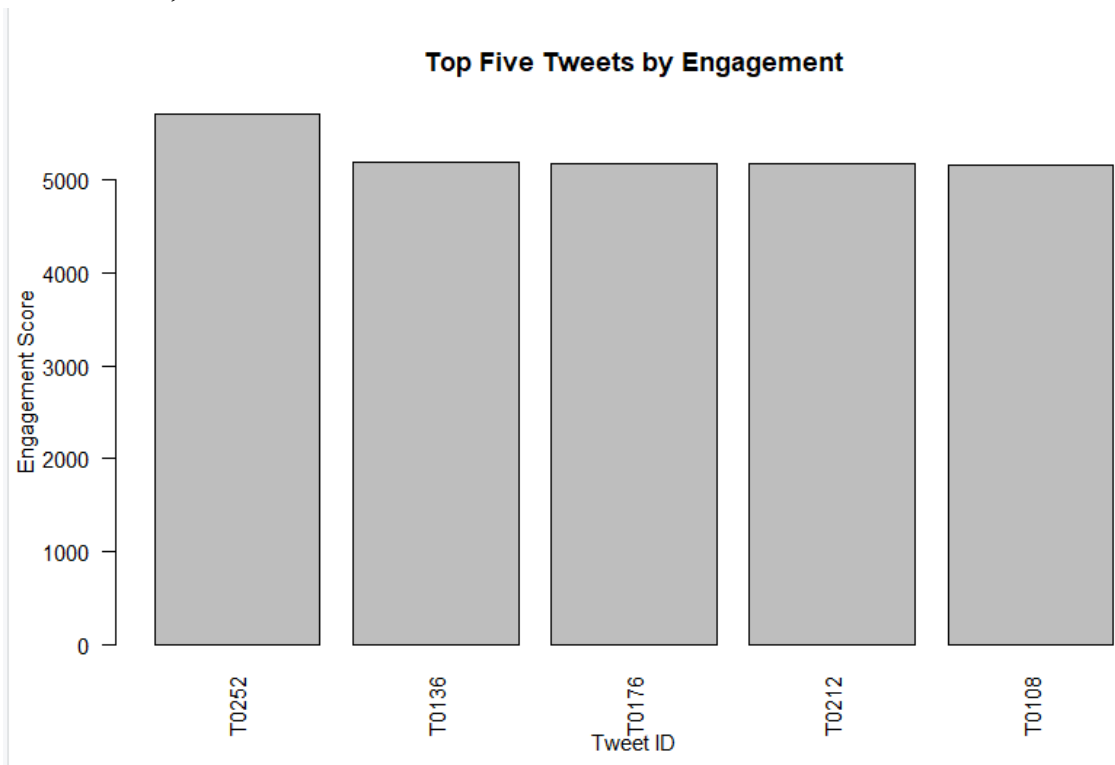
OUTPUT:



E) Top viral tweets;

```
barplot(  
  top_five$Engagement,  
  names.arg = top_five$Tweet_ID,  
  main = "Top Five Tweets by Engagement",  
  xlab = "Tweet ID",  
  ylab = "Engagement Score",  
  las = 2  
)
```

OUTPUT;



FINAL SUMMARY:

```
. /
Total Fake Tweets: 250
>
> cat(
+   "Total Real Tweets:",
+   sum(tweets$Label == "Real"),
+   "\n"
+ )
Total Real Tweets: 250
>
> cat(
+   "Total Fake Retweets:",
+   sum(tweets$Retweets[tweets$Label == "Fake"]),
+   "\n"
+ )
Total Fake Retweets: 233066
>
> cat(
+   "Total Real Retweets:",
+   sum(tweets$Retweets[tweets$Label == "Real"]),
+   "\n"
+ )
Total Real Retweets: 139520
. /
Average Real Retweets: 558.08
>
> cat(
+   "Average Fake Engagement:",
+   fake_engagement,
+   "\n"
+ )
Average Fake Engagement: 3019.96
>
> cat(
+   "Average Real Engagement:",
+   real_engagement,
+   "\n"
+ )
Average Real Engagement: 2093.772
>
> cat(
+   "Highest Fake Propagation Date:",
+   fake_max$Date,
+   "\n"
+ )
Highest Fake Propagation Date: 20468
Highest Fake Propagation: 9434 retweets
>
> cat(
+   "Most Retweeted Tweet:",
+   max_retweet$Tweet_ID,
+   "\n"
+ )
Most Retweeted Tweet: T0250
```